

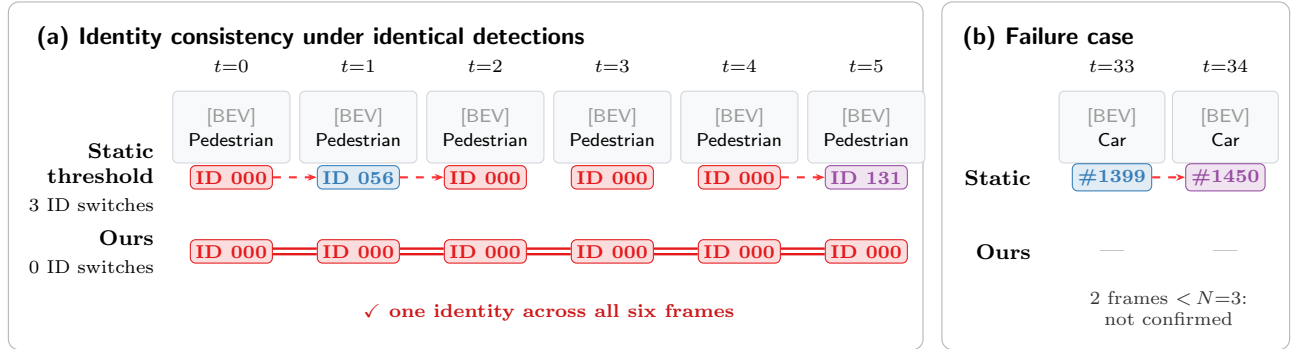


Figure 1: **Identity consistency under identical detections** (nuScenes *val*, sensor frame, detection-budget-matched arms; dashed red arrows mark identity switches). **(a)** One pedestrian in **scene-0925** over six consecutive keyframes. The emitted 3D box is identical in both arms in every frame — same centre, size, orientation and score (0.726–0.833, centre error ≤ 0.27 m, correctly classified throughout) — so the two rows differ only in the identity attached to that box. The static score threshold assigns three distinct identities, returning to the first one in between; confirmation-based initialisation with retrospective emission assigns one. **(b)** The corresponding cost: a correctly classified, well-localised car in **scene-0104** visible for only two keyframes, one short of the $N=3$ confirmation window. The threshold emits it, already split across two identities; our policy does not emit it at all.